

Craftax runs compared: achievements and what unlocks them

8 finished runs, 1.1M env steps each, size50m, one training seed per run. Scored with tools/death_eval.py on the same fixed evaluation worlds. Updated 2026-10-01.

Every number is the mean episode-final achievement count over the evaluation episodes, and each unlock rate is the share of episodes in which that achievement fired. Every run was scored on the same fixed evaluation worlds. Re-scoring the old 5.80 record and the 2.30 run with today's code reproduced both exactly, so the differences below come from the trained policies, not from how they were measured. The non-cheating model (Honest map, trained only on what the agent saw, position dead-reckoned from its own actions) scores 5.87, level with the privileged control's 5.93. Removing the privileged supervision cost nothing measurable.

The control never makes a pickaxe (0 of 30 episodes) and makes a sword in 60%. The honest run, with the same recipe and death fixes, makes both in 20%. So the fixes did not remove the pickaxe; which crafting key a run settles on looks like run-to-run variation or valid-action suppression.

Every map run unlocks WAKE_UP (sleep, then wake) in 83-98% of episodes; vanilla never does. Without it the fixed map model scores 4.20 against vanilla's 4.44; the potential runs stay above vanilla (4.90-5.03).

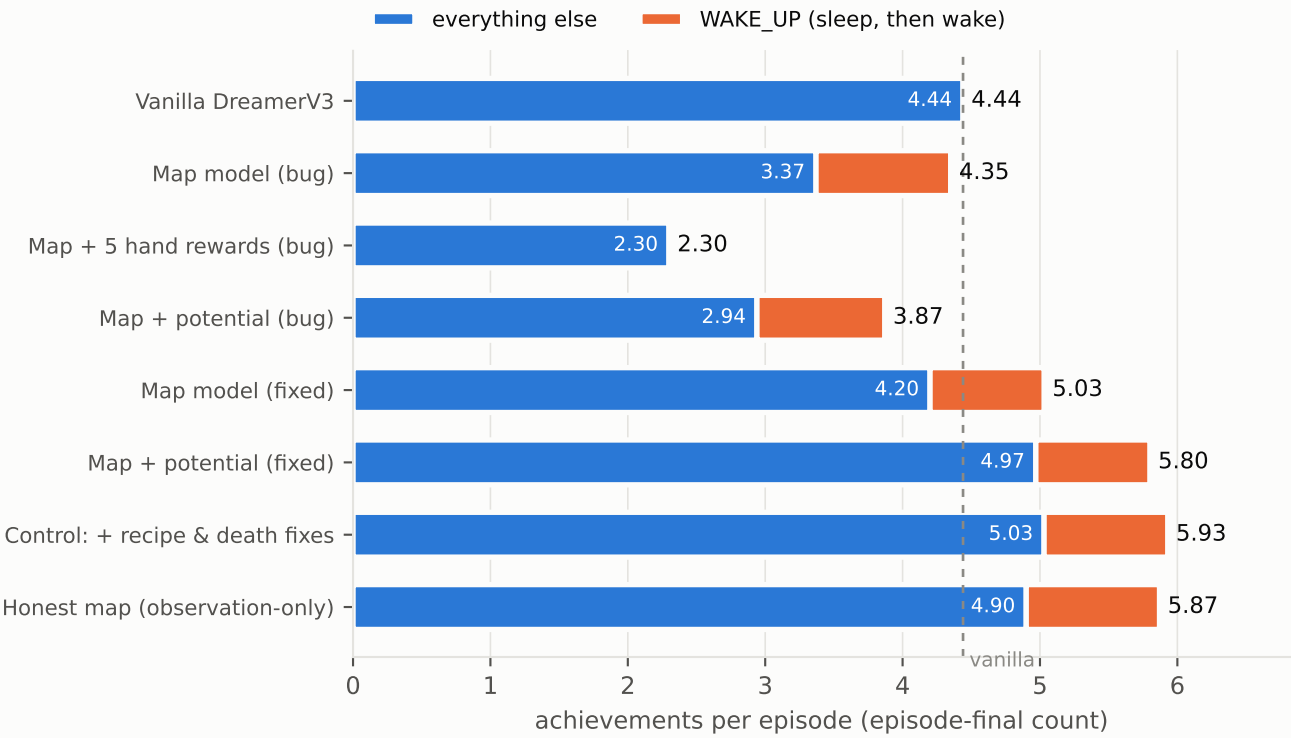
Survival never moved: lifespans 243-294 steps against 261-284 for a random policy. The honest run drinks most reliably of any map run (83%) but lives shortest (243), with half its deaths to mobs or lava.

run	eps	ach.	w/o wake	life	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	25	4.44	4.44	262	64%	24%	0%	12%	80%
Map model (bug)	60	4.35	3.37	294	35%	0%	0%	0%	43%
Map + 5 hand rewards (bug)	50	2.30	2.30	263	0%	0%	0%	0%	44%
Map + potential (bug)	30	3.87	2.94	277	13%	0%	0%	0%	37%
Map model (fixed)	30	5.03	4.20	264	63%	7%	0%	0%	70%
Map + potential (fixed)	30	5.80	4.97	278	90%	30%	10%	20%	60%
Control: + recipe & death fixes	30	5.93	5.03	268	83%	0%	60%	0%	67%
Honest map (observation-only)	30	5.87	4.90	243	87%	20%	20%	7%	83%

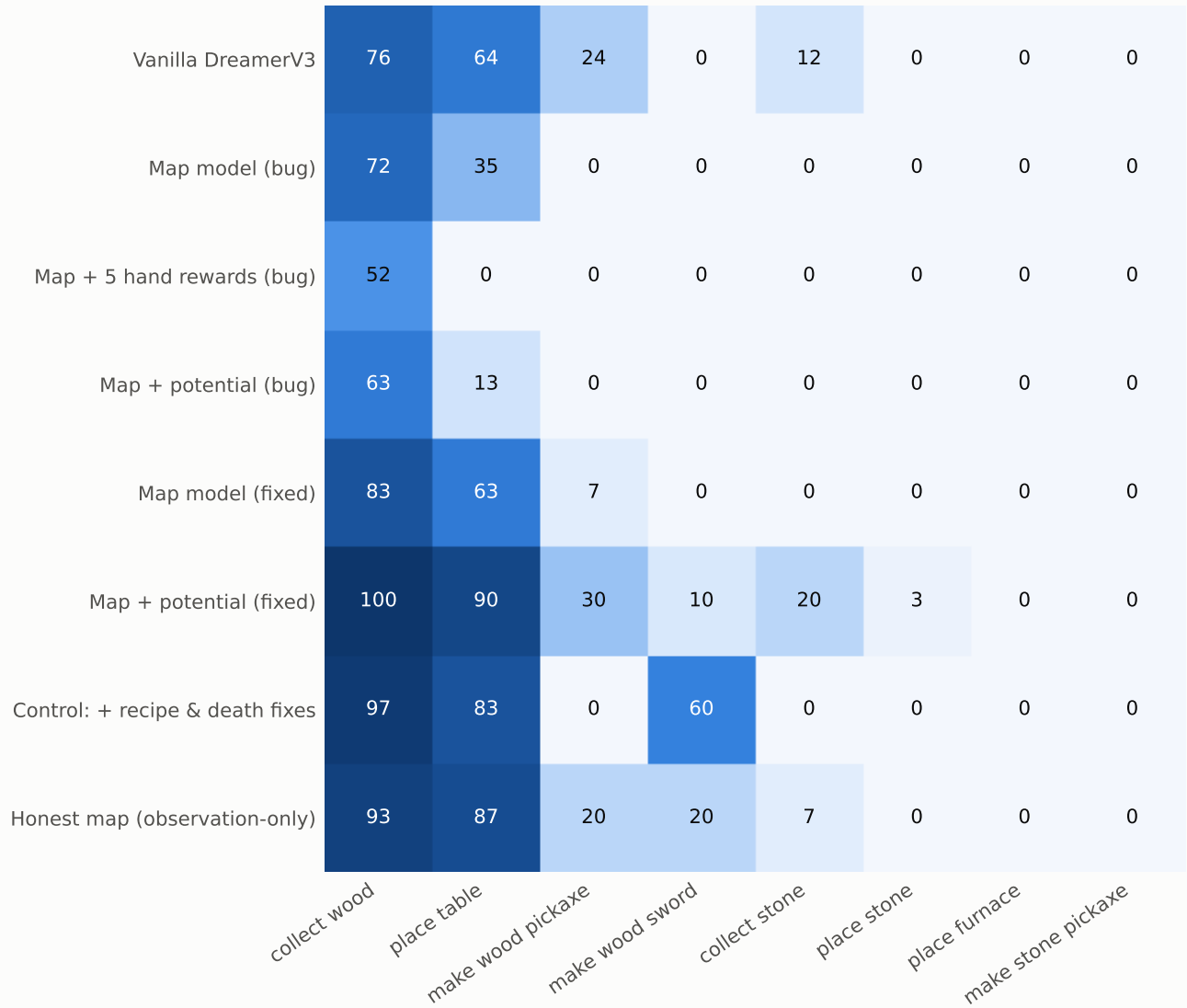
eps = evaluation episodes. ach. = mean episode-final achievements; w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. Rates are % of episodes. 'bug' = imag_shift True, which corrupted the policy gradient (fixed 2026-09-20). map_survive was re-evaluated on 2026-09-28 and reproduced its original 2.30. One training seed per run, so the run-to-run spread is unknown. 'Map + potential (fixed)' is the old 5.80 record; 'Control' adds the recipe and death fixes with the privileged map; 'Honest map' has the same fixes with observation-only targets.

Most of the gain over vanilla is sleep

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order



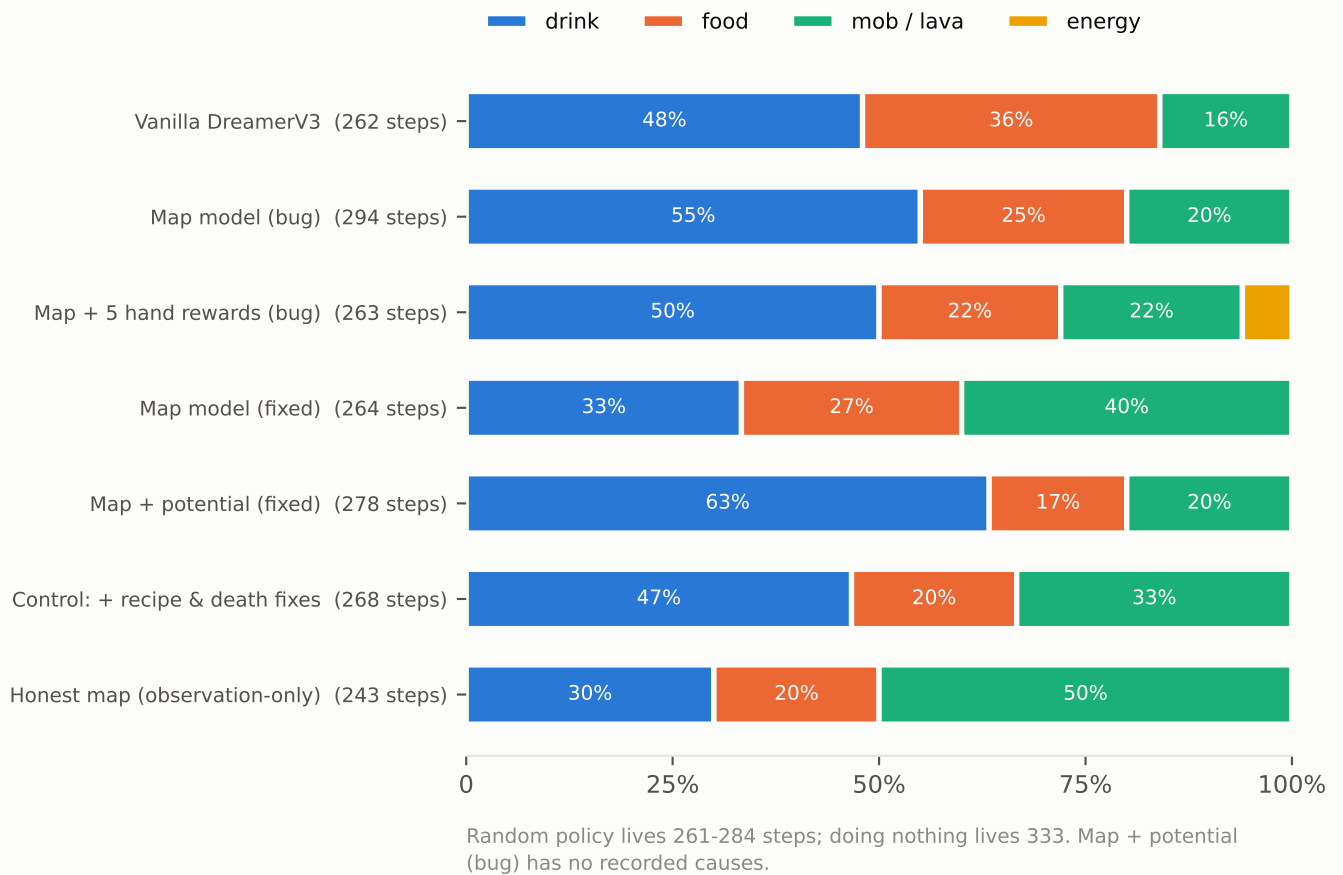
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 54 of 67 were never unlocked by any run.

	vanilla	map, bug	map+5 terms	map+ Φ , bug	map	map+ Φ	map+ Φ +fixes	honest map
collect wood	76	72	52	63	83	100	97	93
place table	64	35	.	13	63	90	83	87
eat cow	4	12	22	13	10	3	17	3
collect sapling	92	92	98	90	93	93	93	90
collect drink	80	43	44	37	70	60	67	83
make wood pickaxe	24	.	.	.	7	30	.	20
make wood sword	.	.	.	.	.	10	60	20
place plant	92	77	.	73	93	87	83	87
defeat zombie	.	5	12	3	.	.	3	.
collect stone	12	.	.	.	.	20	.	7
place stone	.	.	.	.	.	3	.	.
defeat skeleton	.	2	2	.	.	.	.	.
wake up	.	98	.	93	83	83	90	97

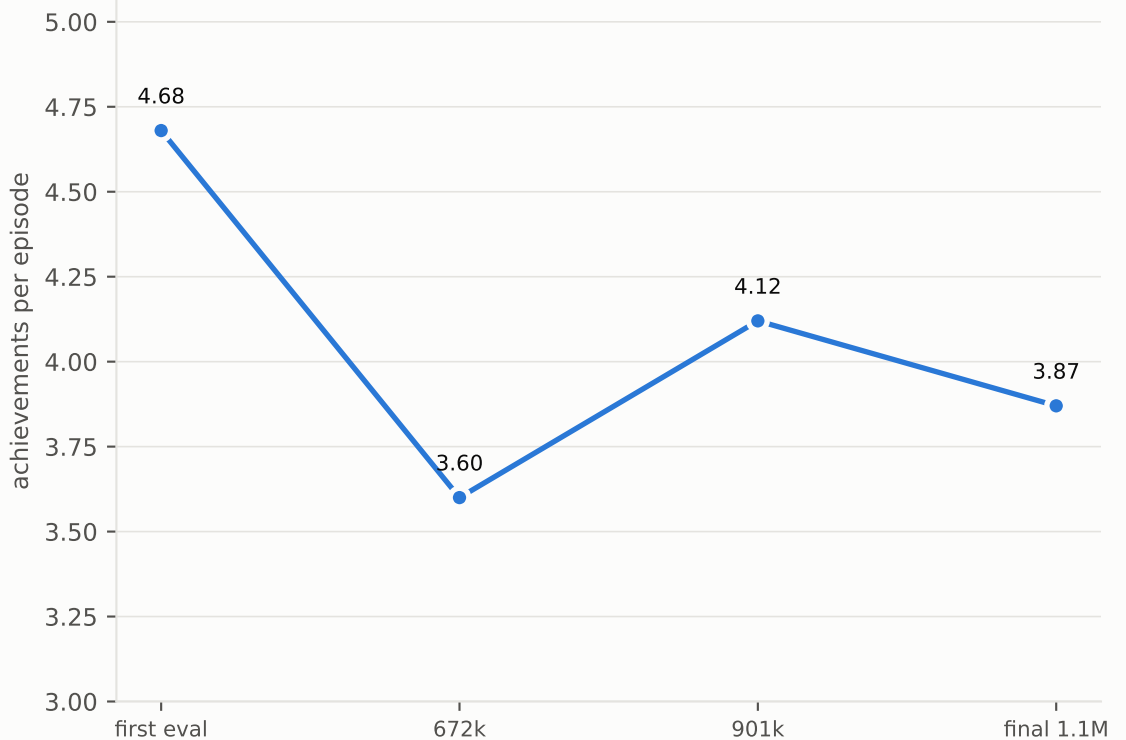
Survival never moved

Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



What the results say, and what would settle the rest

1. Honest training costs nothing measurable

Honest map 5.87 vs privileged control 5.93: inside the noise of single runs (one run moved 1.08 between its own checkpoints). Same with WAKE_UP removed: 4.90 vs 5.03. The map no longer learns from terrain the agent never saw, and the score did not drop.

2. The pickaxe: the fixes are probably not the cause

Control: pickaxe 0%, sword 60%. Honest map, same recipe and death fixes: pickaxe 20%, sword 20%. Old record, without the fixes: pickaxe 30%, sword 10%. If the death fix had removed the pickaxe's incentive, the honest run should have lost it too. That leaves (a) run-to-run variance and (b) valid-action suppression -- both crafts are valid in the same state, and which key survives looks close to arbitrary per run. The honest run also differs from the control in its map targets, so this weakens rather than rules out (c).

3. The map model's gain is mostly sleep

WAKE_UP fires in 83-98% of episodes for every map run that learned to sleep, and 0% for vanilla. Remove it and the map model alone (4.20) is below vanilla (4.44). The potential is what adds tech progress. Why map runs learn to sleep is an open question; it is systematic, not seed noise.

4. Survival is still unsolved

Every run lives 243-294 steps, inside or below the random band. The honest run drinks most reliably (83%) yet dies earliest, half the time to mobs or lava. Thirst is no longer its main killer; combat is.

5. What to run, cheapest first

Experiment A: probability of each crafting key in table-with-wood states, on the control, the honest run and the old record (no training). Then second seeds, which are the only way to separate (a) from (b) and what reviewers expect. Explain the WAKE_UP gap before claiming the map model helps.